

Methods and Determination of Quadrupole β_2 Deformation Parameters Across Experimental Techniques

Parker Lewis

November 18, 2025

Abstract

Contents

Abstract	1
1 Introduction	3
2 Theory	3
2.1 Nuclear Structure and the Shell Model	3
2.2 Beyond the Shell Model: Collectivity and Deformation	3
2.3 Quadrupole Moments and the β_2 Deformation Parameter	3
2.4 Relation Between $B(E2)$, Q_0 , and β_2	3
3 Experimental Techniques	3
3.1 Electron Scattering and Fixed-Target Measurements	3
3.1.1 Low-Energy Coulomb Excitation	3
3.2 Heavy-Ion Collision and Reaction-Based Methods	3
4 Methodology for Extracting β_2	3
5 Survey of Results from Recent Literature	3

1 Introduction

2 Theory

2.1 Nuclear Structure and the Shell Model

2.2 Beyond the Shell Model: Collectivity and Deformation

2.3 Quadrupole Moments and the β_2 Deformation Parameter

2.4 Relation Between $B(E2)$, Q_0 , and β_2

3 Experimental Techniques

3.1 Electron Scattering and Fixed-Target Measurements

3.1.1 Low-Energy Coulomb Excitation

3.2 Heavy-Ion Collision and Reaction-Based Methods

4 Methodology for Extracting β_2

5 Survey of Results from Recent Literature

6 Conclusion